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1

Properties

Various models for the thermophysical properties of different materials will be implemented. This includes

- Water
- Lead
- Sodium
- UO₂
- Metallic Uranium
- Inconel
- Zircalloy
- Various gases

1.1 Water Properties

The water properties are implemented via the IAPWS95 module, which enables pytorch ported parallelized solutions to the 1995 IAPWS equation of state (EOS) along with the IAPWS releases for thermal conductivity and viscosity.

1.1.1 IAPWS 1995 Equation of State:

1.1.2 IAPWS 1997 Equation of State:

1.1.3 IAPWS Thermal Conductivity and Viscosity:

1.2 Liquid Metal Properties

The properties of liquid metals are implemented in the LiqProps module, primarily recommended equations from IAEA recommendations of Sobolev and his later paper (correcting some errors). These assume atmospheric pressure, which yields little error in reactor conditions unless under accident conditions.

1.3 UO₂

Uranium oxide fuel has models for thermal conductivity, thermal expansion, swelling, and emissivity.

1.3.1 Thermal conductivity: Two models are available for the thermal conductivity of UO₂ fuel. These include the classic model of Klimenko, which relies solely on temperature, and the MatLib formulation, which includes burnup and gadolinia concentrations. The corresponding conductivity integrals are also given

The Klimenko-Zorin model uses the temperature in kelvin normalized over 1000, $\tau = T \cdot 10^{-3}$. It assumes a fuel density of 95 percent. It has uncertainty of 10 percent from 298K to 2000K and 20 percent in the range 2000-3120K. The conductivity is given as:

$$k_f(T) = \frac{100}{7.5408 + 17.692\tau + 3.6142\tau^2} + \frac{6400}{\tau^{\frac{5}{2}}} \exp\left(\frac{-16.35}{\tau}\right)$$

The matpro model is given by:

$$k_{95} = \left(\frac{1}{A + \alpha gad + BT + f(Bu) + (1 - 0.9e^{-0.04Bu})g(Bu)h(T)} \right) + \frac{C}{T^2} \exp\left(-\frac{D}{T}\right) \quad (1)$$

$$h(T) = \frac{1}{1 + 396 \exp(-Q/T)} \quad (2)$$

Where,

- k_{95} = Thermal conductivity of 95% TD fuel, W/m-K
- T = Temperature, K
- Bu = Burnup, GWd/MTU
- $f(Bu)$ = Effect of fission products in crystal matrix (solution) = $0.00187Bu$
- $g(Bu)$ = Effect of irradiation defects = $0.038Bu^{0.28}$
- $h(T)$ = Temperature dependence of annealing on irradiation defects (Equation 2)
- Q = Temperature-dependent parameter ("Q/R") = 6380 K
- A = 0.0452 m-K/W
- B = 2.46×10^{-4} m-K/W/K
- C = 3.5×10^9 W-K/m
- D = 16361 K
- α = Constant = 1.1599
- gad = Weight fraction of gadolinia, unitless

The input ranges are given as:

- Gd Content: $0 \leq f_{Gd} \leq 10$
- Temperature: $T \in [300, 2800]$
- Burnup: $Bu_{UO_2} \in [0, 90], Bu_{UO_2-Gd} \in [0, 50]$ GWd/MTU

1.3.2 Emissivity: The emmissivity model for UO2 is the same as MatLib, given by:

$$\epsilon = 0.78557 + 1.5263 \times T \quad (3)$$

Where T is in Kelvin. Its applicability and uncertainty:

- Fuels: UO2, MOX, UO2-Gd
- Gadolinia content: $0\% \leq f_{Gd} \leq 10\%$
- Temperature: $300K \leq T \leq 2500K$
- Uncertainty: $\sigma=0.072$

1.3.3 Thermal Expansion: The model for thermal expansion is the same as in matlib, given by:

$$\frac{\Delta L}{L} = (9.80 \cdot 10^{-6}) \cdot T - (2.61 \cdot 10^{-3}) + (3.16 \cdot 10^{-1}) \exp\left(\frac{1.32 \cdot 10^{-19}}{kT}\right) \quad (4)$$

1.3.4 Swelling:

1.4 *Steels*

Correlations for austenitic

1.4.1 HT9 Steel Models:

1.5 *Gas*

1.5.1 Thermal Conductivity: A correlation for the thermal conductivity of various gases is identical to MatLib

1.6 *Future materials*

It is intended in the future to add a module for the EOS of CO₂ along with some property correlations for various gaseous coolants for HTGRs

2

Heat Transfer Coefficient Models

The HTC module includes several classes for different heat transfer coefficient correlations for various working fluids. This includes single phase water, saturated water, supercritical water, lead, and sodium metal coolants. Current support will focus principally on supercritical water. Support for other coolants may be added later.

2.1 *Single Phase Water*

This class includes the following heat transfer correlations for single phase water:

- Dittus Boelter
- Petukhov
- Gnielinski

2.1.1 Dittus Boelter:

2.1.2 Petukhov:

2.1.3 Gnielinski:

2.2 *Two Phase Water*

The following two phase heat transfer coefficients for subcooled and saturated water are supported:

- Schrock and Grossman
- Chen
- Bjorge

2.2.1 Schrock and Grossman:

2.2.2 Chen:

2.2.3 Bjorge:

2.3 Supercritical Water

The following htc correlations are implemented for the heat transfer coefficient of supercritical water:

- Swenson
- Chen

2.3.1 Swenson Correlation:

2.3.2 Chen et al:

2.4 Liquid Sodium

2.4.1 Notter-Sleicher correlation:

2.5 Liquid Lead and Lead-Bismuth Eutectic

2.5.1 Shen:

3

Friction Models

The FRICT module contains friction factor models for various working fluids.

3.1 Single Phase Water

3.1.1 McAdams:

3.1.2 Colebrook:

3.2 Supercritical Water

Both the Filonenko and Wu correlations are implemented in the f_SCW class. The Wu correlation is corrected for rod bundle geometry.

3.2.1 Filonenko Correlation:

3.2.2 Wu Correlation for Rod Bundles:

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Fuel Pin Heat Transfer Models/Routines

The heat transfer model PinHT, includes several equations and models for modeling heat transfer in nuclear fuel pins

4.1 Gap Heat Transfer

Gap heat transfer is modeled using the `htc_gap` function. This function approximates the heat transfer coefficient of an open gas gap (ignoring closure effects). It assumes the diametral gas gap is much less than the radius of the fuel/inner clad (ignoring the radial effects of heat conduction). It takes arguments of outer fuel temperature (T_{fo}), inner clad temperature (T_{ci}), the diametral gap width (δ), the conductivity function for the gas (k_{gas}), as well as the cladding and fuel emissivities (ϵ_c , ϵ_f). It returns the heat transfer coefficient of the gap.

The model is:

$$T_{ave} = \frac{T_{fo} + T_{ci}}{2}$$
$$k = k_{gas}(T_{ave})$$

The conductive contribution to the gap heat transfer coefficient is approximated as simple planar conduction across the gap width δ :

$$h_{cond} = \frac{k}{\delta}$$

The radiative contribution treats the fuel outer surface and clad inner surface as two infinite parallel gray surfaces, giving an effective radiation heat transfer coefficient:

$$h_{rad} = \frac{\sigma (T_{fo}^4 - T_{ci}^4)}{(\epsilon_f^{-1} + \epsilon_c^{-1} - 1) (T_{fo} - T_{ci})}$$

where σ is the Stefan-Boltzmann constant.

The total gap heat transfer coefficient is the sum of the conductive and radiative contributions:

$$h_{gap} = h_{cond} + h_{rad}$$

The result is returned in units of $\text{W m}^{-2} \text{K}^{-1}$ by default, with an optional scale factor (`units`) to convert the energy unit to kJ or MJ.

4.2 Clad conduction

The conduction across the cladding is computed using the `T_ci` function, which computes the inner cladding temperature. It takes arguments of outer clad radius (R_{co}), inner clad radius (R_{ci}), cladding conductivity (k_c), outer cladding temperature (T_{co}), and the LHGR (q). It is computed as:

$$T_{ci} = T_{co} + \frac{q'(z)}{2\pi k_c} \log \left(\frac{R_1}{R_2} \right)$$

Where R_1 is the larger of the two input radii (R_{co} may be smaller than R_{ci} in the inner region of an annulus)

4.3 Cylindrical Heat transfer

This function resolved the steady state radial heat equation for a cylinder. It uses a numerical solver to compute the temperature at a given radial position. It is formulated as:

$$\int k_f(T) dT = \frac{q'''(z)}{4}r^2 + C_1 \log(r) + C_2$$

which is solved iteratively for T given the radius. Functions are also available to compute the heat flux and LHGR at a specified radius.

4.4 Annular Heat Equation

This function represents the solution to the heat equation with an annular heat source. It takes arguments of the inner and outer fuel temperature, the volumetric heat generation rate, a conductivity integral function, and the solution constants. It returns the inner and outer heat fluxes. The inner heat flux has a sign notation to be positive when moving in the negative radial direction.

4.5 Rod Bundle Correction factors

The Weissman and Presser equations are available in the Bundle class.

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Monte Carlo Subroutine and Error Propagation
